Category

Company Analysis: Forecasting

FORECASTING OPERATING EXPENSES AND WORKING CAPITAL

Learning Outcome

- explain approaches to forecasting a company's operating expenses and working capital

Issuers' disclosures about operating costs are typically less detailed than revenue disclosures. Rather than modeling costs separately for different geographic regions, business segments, or product lines, analysts are often forced to use more aggregated forecast objects such as consolidated financial statement lines (e.g., cost of sales, SG&A) or summary measures like EBITDA margins on a consolidated or segment basis. The forecasts for revenues and costs should still be coherent. If sales of a relatively low-margin product, segment, or geography are forecast to grow faster than other revenues, some level of overall profit margin deterioration should be forecast, even if the analyst is uncertain about the precise margins earned on each object. For example, food and grocery typically has low gross profit margins, so an analyst of a grocery chain that also sells higher-margin items, such as alcoholic products or pharmaceutical products, would want to forecast any change in product mix sold.

Cost of Sales and Gross Margins

Cost of sales (cost of goods sold, or COGS) is typically the single largest cost for companies that make and/or sell products. Because it has a direct link with sales, forecasting this item as a percentage of sales (or as a gross margin) is usually a good approach. If a company is losing market share to new substitute products with lower prices, gross margins are likely to decline. But if the company is gaining market share because it has introduced differentiated products—especially if it has done so in combination with achieving cost advantages—gross margins are likely to rise.

Cost of sales is typically a large cost, so even a single basis point in a gross margin forecast can have a material impact on forecasts of operating profit and free cash flow. Analysts should consider whether a more detailed analysis of these costs (e.g., by segment, input, or product line, or by volume and price components) is possible to better justify the forecast. For example, some companies face fluctuating input costs that can be passed on to customers only with a time lag. Particularly for companies with low gross margins, sudden shocks in input costs can affect operating profit significantly. A good example is the sensitivity of airlines' profits to unhedged changes in jet fuel costs. In these cases, a breakdown of both costs and sales into volume and price components is essential for developing short-term forecasts, even if analysts use the overall relationship between sales and input costs for developing longer-term forecasts, as shown in Exhibit 3.

Exhibit 3:

The Effect of Prices and Costs on Gross Profit and Margin

Assume that a company's COGS as a percentage of sales equals 25% and that the quantity sold is the same in Period 2 as in Period 1. If input costs double in Period 2 and the company can pass the entire increase on to its customers through a 25% price increase, COGS as a percentage of sales will increase (to 40%) because an equal absolute amount has been added to the numerator and to the denominator.

Gross profit	75.0	75.0
COGS as % of sales	25%	40%
Gross margin %	75%	60%

Thus, although the absolute amount of gross profit will remain constant, the gross margin will decrease (from 75% to 60%).

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Analysts should incorporate a company's hedging strategy into a forecast. For example, commodity-driven companies' gross margins almost automatically decline if input prices of needed commodities increase significantly, because the variable costs increase at a faster rate than output prices. Through hedging strategies, a company can mitigate the impact on profitability. For example, brewers often hedge the cost of barley, a key raw material needed for brewing beer, one year in advance. Although companies usually do not disclose specific hedging positions, their hedging strategy is often disclosed in the notes to the financial statements. Further, the negative impact of increasing sales prices on sales volume (if product demand is price elastic) can be mitigated by a policy of gradual sales price increases. For example, if the brewer expects higher barley prices because of a poor harvest, the brewer can slowly increase prices to avoid a strong price jump next year.

While competitors' gross margins can provide a useful cross-check for forecasting gross margin, the margins may not be comparable given differences in business models. For example, some retailers own and operate their own stores, whereas other companies operate as wholesalers with franchised retail operations. In the franchise model, most of the operating costs are incurred by the franchisee; the wholesaler sells products with only a small markup to these franchisees. Compared with a retailer with its own stores, a wholesaler will have a much lower gross margin but also much lower operating costs.

SG&A Expenses

SG&A expenses are the other main type of operating costs. In contrast to cost of sales, SG&A expenses often have a less direct relationship with revenues. As an illustration of the profit impact of COGS and SG&A, consider the case of a Thai cement and building materials company, Siam Cement Group, from 2017 to 2018. A summary of the company's key income statement items is shown in Exhibit 4.

Exhibit 4:

Siam Cement Group 2017–18 Financials

	2017 (THB billions)	2018 (THB billions)	YoY%	Percentage of Sales 2017	Percentage of Sales 2018
Net sales	450.92	478.44	6.1	100.0	100.0
COGS	349.31	383.46	9.8	77.5	80.1
Gross profit	101.61	94.98	−6.5	22.5	19.9
SG&A	52.58	55.09	4.8	11.7	11.5
Selected SG&A items:					
Salary and personnel expenses	24.24	23.98	−1.1	5.4	5.0
Freight costs	11.63	11.55	−0.7	2.6	2.4
Research and development	4.18	4.67	11.7	0.9	1.0
Promotion and advertising	2.62	2.58	−1.5	0.6	0.5
Operating income	49.03	39.89	−18.6	10.8	8.4

Note: "YoY%" means year-over-year percentage change.
Source: Based on information in Siam Cement Group's annual reports.

operating costs, SG&A expenses grew 4.8%, declining slightly as a percentage of revenue. Operating income fell 18.6%. This experience contrasts with the company's experience in 2016 (not shown), when lower input costs resulted in a widening of the gross margin by 240 basis points versus the prior year.

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Although SG&A expenses overall are generally less closely linked to revenue than is cost of sales, certain SG&A expenses could be more variable than others. Specifically, selling and distribution expenses often have a large variable component and can be modeled as a percentage of sales, while general corporate costs are more fixed and might be better modeled using a fixed growth rate based on expected wage inflation.

Segment disclosures often include operating and EBITDA margin profitability by segment, but generally do not include cost items such as cost of sales, SG&A, etc., by segment. If an analyst is constructing a model based on segment forecasts, the analyst might instead use summary measures on a segment basis, as shown in Example 1.

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EXAMPLE 1

Analysis of a Consumer Goods Company with Multiple Segments

A consumer goods company reported an overall underlying operating margin of 19.1% in 2X19. As shown in Exhibit 5, the operating margin is lowest in its fastest-growing segment, Home Care. The other two segments enjoy higher margins but are growing more slowly.

Exhibit 5:

Segment Revenue and Profits (EUR millions, unless noted)

Segment	2X19	2X18	'19/'18 YoY	Avg. Growth Rate, 2X17–2X19
Beauty & Personal Care	21,868	20,624	6%	3%
Foods & Refreshments	19,287	20,227	−5%	−5%
Home Care	10,825	10,131	7%	3%
Total revenues	51,980	50,982	2%	0%
Underlying operating profit				
Beauty & Personal Care	4,960	4,543	9%	
Foods & Refreshments	3,382	3,576	−5%	
Home Care	1,605	1,344	19%	
Total	9,947	9,463	5%	
Underlying operating profit margin				
Beauty & Personal Care	22.7%	22.0%		
Foods & Refreshments	17.5%	17.7%		
Home Care	14.8%	13.3%		
Total	19.1%	18.6%		

Note: Underlying operating profit is a non-IFRS operating profit measure, equal to IFRS operating profit adjusted for items such as restructuring costs.

Source: Example adapted from Unilever.

1. Determine the estimated sales, operating profit, and operating profit margin by using the following two approaches: (A) Assume total sales growth of 2.0% and overall underlying operating margin of 19.1% for the next five years, and (B) assume each individual segment's sales growth and underlying operating margin

Solution:

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Exhibit 6 shows that operating profit after five years will be EUR10,962 million under Approach A and EUR11,549 million under Approach B.

Exhibit 6:

Sales and Operating Profit, 2X19–2X24E (EUR millions, unless noted)

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Approach A	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Total revenues	51,980	53,020	54,080	55,162	56,265	57,390
Growth rate		2%	2%	2%	2%	2%
Underlying operating profit	9,947	10,127	10,329	10,536	10,747	10,962
Growth rate		2%	2%	2%	2%	2%
Underlying operating profit margin	19.1%	19.1%	19.1%	19.1%	19.1%	19.1%

Approach B Sales	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	21,868	23,187	24,586	26,069	27,641	29,308
Growth rate		6%	6%	6%	6%	6%
Foods & Refreshments	19,287	18,391	17,536	16,721	15,944	15,203
Growth rate		−5%	−5%	−5%	−5%	−5%
Home Care	10,825	11,567	12,359	13,205	14,110	15,077
Growth rate		7%	7%	7%	7%	7%
Total revenues	51,980	53,144	54,481	55,995	57,695	59,588

Margins	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	22.7%	22.7%	22.7%	22.7%	22.7%	22.7%
Foods & Refreshments	17.5%	17.5%	17.5%	17.5%	17.5%	17.5%
Home Care	14.8%	14.8%	14.8%	14.8%	14.8%	14.8%

Underlying operating profit	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	4,960	5,259	5,576	5,913	6,269	6,648
Foods & Refreshments	3,382	3,225	3,075	2,932	2,796	2,666
Home Care	1,605	1,715	1,832	1,958	2,092	2,235
Total	9,947	10,199	10,484	10,803	11,157	11,549

2. Compare and explain the results under the two alternative approaches (A and B) in Question 1.

Hide Solution

Solution:

Approach A assumes a constant 2% total sales growth rate and a stable 19.1% underlying operating margin. Thus, the operating profit growth rate is in line with the revenue growth rate and constant at 2%, which therefore assumes no difference in growth rates and profitability of the segments. Approach B assumes growth rates of 6%, −5%, and 7% of sales for the

underlying operating profit margin of 6 bps due to the mix effect of different segment margins. In 2024E, Approach A yields an underlying operating profit margin of 19.1%.

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3. Assume the company can grow segment revenues over the next five years at the following rates: Beauty & Personal Care, 3.0%; Foods & Refreshments, 2.0%; Home Care, 4.0%. But underlying operating profit margins in Beauty & Personal Care will fall 20 bps annually for the next five years (because of high competition, limited growth, and costs resulting from the adoption of sustainable packaging), and operating profit margins in the Foods & Refreshments and Home Care segments will increase by 15 and 50 bps, respectively, each year for the next five years (helped by increasing demand for the company's products and better utilization of its factories). Calculate the overall underlying operating profit margin in each of the next five years.

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Hide Solution

Solution:

As shown in Exhibit 7, the overall underlying operating profit margin improves from 19.1% in 2X19 to 19.5% in 2X24, because the margin decline in Beauty & Personal Care is more than offset by the margin increase in Foods & Refreshments and the faster-growing Home Care segment.

Exhibit 7:

Sales and Operating Profit, 2X19–2X24E
(EUR millions, unless noted)

Sales	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	21,868	22,524	23,200	23,896	24,613	25,351
Growth rate		3%	3%	3%	3%	3%
Foods & Refreshments	19,287	19,673	20,066	20,468	20,877	21,294
Growth rate		2%	2%	2%	2%	2%
Home Care	10,825	11,258	11,708	12,177	12,664	13,170
Growth rate		4%	4%	4%	4%	4%
Total revenues	51,980	53,455	54,974	56,540	58,153	59,816

Margins	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	22.7%	22.5%	22.3%	22.1%	21.9%	21.7%
Foods & Refreshments	17.5%	17.7%	17.8%	18.0%	18.1%	18.3%
Home Care	14.8%	15.3%	15.8%	16.3%	16.8%	17.3%

Underlying operating profit	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	4,960	5,064	5,169	5,277	5,386	5,496
Foods & Refreshments	3,382	3,479	3,579	3,681	3,786	3,894
Home Care	1,605	1,725	1,853	1,988	2,131	2,282
Total	9,947	10,268	10,601	10,946	11,303	11,672
Margin	19.1%	19.2%	19.3%	19.4%	19.4%	19.5%

Working Capital Forecasts

liabilities, as demonstrated in Example 2.

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EXAMPLE 2

Working Capital Forecasts with Efficiency Ratios

Exhibit 8 shows revenue, COGS, and year-end working capital account balances for YY Ltd., a fictional company, for years 1–3. Based on the data in the exhibit, answer Questions 1–3.

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Exhibit 8:

YY Ltd. Financial Data (CNY millions)

Year	1	2	3
Revenue	174,915	205,839	245,866
COGS	152,723	177,285	209,114
Accounts receivable	5,598	6,949	10,161
Inventory	29,481	32,585	41,671
Accounts payable	46,287	59,528	72,199

1. Calculate days sales outstanding (DSO), inventory days on hand (DOH), and days payable outstanding (DPO) for years 1, 2, and 3, using year-end balances and assuming a 365-day fiscal year.

Hide Solution

Solution:

Recall that days sales outstanding is equal to accounts receivable/(revenues/365), inventory days on hand is equal to inventories/(COGS/365), and days payable outstanding is equal to accounts payable/(COGS/365). Using the data in Exhibit 8, the three ratios for years 1–3 are as follows:

Year	1	2	3
Days sales outstanding	12	12	15
Inventory days on hand	70	67	73
Days payable outstanding	111	123	126

2. Your colleague Liang forecasts revenue growth of 18%, 16%, and 13% and gross margins of 17%, 17%, and 16% in years 4, 5, and 6, respectively, for YY Ltd. Using Liang's forecasts, calculate expected revenue and COGS in years 4, 5, and 6.

Hide Solution

Solution:

Using Liang's forecasts for annual revenue growth and gross margins and the data in Exhibit 8, expected revenue and COGS for years 4–6 are as follows:

Year (CNY millions)	1	2	3	4	5	6
Revenue	174,915	205,839	245,866	290,122	336,541	380,292
Growth rate		18%	19%	18%	16%	13%

Gross margin	13%	14%	15%	17%	17%	16%

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3. Liang forecasts that days sales outstanding, inventory days on hand, and days payable outstanding in years 4, 5, and 6 will remain the same as the year 3 amounts. Using Liang's forecasts as well as forecast revenue and COGS, calculate expected accounts receivable, inventories, and accounts payable year-end balances for years 4, 5, and 6.

Hide Solution

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Solution:

Each of the efficiency ratios can be rearranged to yield working capital balances, because we have values for two of the three variables in them: the efficiency ratios are assumed to remain constant from year 3 levels, and revenue and COGS have already been forecast.

Days sales outstanding is equal to accounts receivable/(revenues/365); thus, accounts receivable is equal to days sales outstanding × (revenues/365). Similarly, inventories is equal to inventory days on hand × (COGS/365), and accounts payable is equal to days payable outstanding × (COGS/365).

Using the data in Exhibit 8 and the revenue and COGS forecasts, year-end working capital account balances are as follows:

Year (CNY millions)	1	2	3	4	5	6
Revenue	174,915	205,839	245,866	290,122	336,541	380,292
COGS	152,723	177,285	209,114	240,801	279,329	319,445
Accounts receivable	5,598	6,949	10,161	11,990	13,908	15,716
Inventories	29,481	32,585	41,671	47,985	55,663	63,657
Accounts payable	46,287	59,528	72,199	83,139	96,442	110,292
Days sales outstanding	12	12	15	15	15	15
Inventory days on hand	70	67	73	73	73	73
Days payable outstanding	111	123	126	126	126	126

While a historical results approach is common for working capital efficiency ratios, analysts can also use the other forecast approaches. Exhibit 9 discusses forecast objects and approaches that Elaine Nguyen can use for Warehouse Club Inc.'s operating costs and working capital.

Exhibit 9:

Forecast Approaches for Warehouse Club Inc.'s Operating Costs and Working Capital

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Gross margin (on net sales)	Historical results	From 2X10 to 2X19, Warehouse's gross margin increased by an average of 2 bps per year. Such an increase may reflect increasing bargaining

		gross margin as a forecast.
	Historical base rates and convergence	Both the industry average gross margin and that of larger, more mature retailers such as Walmart are much greater than Warehouse's gross margin. But Warehouse's business model is to use a low gross margin on net sales to drive high-margin membership revenues. Therefore, Nguyen probably will not use this approach.
	Management guidance	If Warehouse management provides guidance on margins, Nguyen can use this approach as the forecast, after evaluating management's forecasting credibility.
	Analyst's discretionary forecast	Nguyen may formulate a discretionary forecast for gross margin by exploring the relationship between it, inflation/deflation in Warehouse's product categories, and management's desire for member savings vs. competitors to equal 5× the annual price of membership.

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Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Selling, general, and administrative expenses (SG&A) as a percentage of sales	Historical results	SG&A, which captures the cost of operating stores and some merchandise distribution, is primarily composed of employee payroll. Relative to sales, it has remained mostly stable at or around 9%. Nguyen may forecast it to remain the same.
	Historical base rates and convergence	Warehouse's SG&A as a percentage of sales is already lower than that of larger retailer competitors (and substantially lower than the retail industry average), owing to its business model differences that emphasize store productivity and cost leadership. Nguyen will therefore probably not use this approach to forecast SG&A.
	Management guidance	If Warehouse management provides guidance on SG&A, Nguyen can use this approach as the forecast, after evaluating management's forecasting credibility.
	Analyst's discretionary forecast	Since SG&A is primarily composed of employee payroll, Nguyen may forecast it based on projected employee productivity (e.g., employees to stores) and projected wage rates, which may be based on macroeconomic variables like projected economy-wide wage growth.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Working capital ratios (e.g., DSO, DOH, DPO)	Historical results	As of 2X19, Warehouse's DSO, DOH, and DPO were 3, 30, and 28, respectively. DSO and DPO have been stable for the last 10 years, while DOH has decreased by 7 days since 2X10. Nguyen might forecast a stable DSO and DPO and a modestly declining DOH over the next several years.
	Historical base rates and convergence	Warehouse's working capital ratios and cash conversion cycle are materially better than the industry average and some larger peers and have been for many years, owing to its business model differences. Nguyen will therefore probably not use this forecast approach, unless Warehouse changes its business model (e.g., it launches an e-commerce business).

		working capital to sales ratio), Nguyen can use this approach as the forecast, after evaluating management's forecasting credibility.
	Analyst's discretionary forecast	Nguyen may develop a discretionary forecast for working capital by, for example, estimating net working capital to sales ratios by product category (e.g., fresh foods, packaged foods, non-food merchandise) and applying them to a net sales forecast also broken down by category.

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Question Set

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Iliso Marketplace Ltd., the e-commerce retailer introduced in prior modules, reports its GMV by product category: in 2X19, the mix was 55% food and grocery, 20% media and electronics, 8% home goods and décor, and 17% apparel and beauty products. However, Iliso presents only aggregated financial results (no segment disclosures).

Iliso's cost of revenue is primarily composed of the inventory costs of products sold to customers where Iliso records revenue gross. Inventory costs include shipping and handling costs. Additionally, cost of revenue includes outbound shipping and logistics-related expenses, as well as delivery service costs from the third-party merchant business, where Iliso is the delivery service provider. Cost of revenue also includes depreciation expense related to inventory handling and shipping assets.

Iliso's operating, general, and administrative expenses include costs incurred in operating and staffing its fulfillment centers, customer service costs, payment processing fees, costs related to the design, execution, and maintenance of Iliso's technology infrastructure, and general corporate function costs.

Iliso has minimal accounts receivable, because customers pay with credit cards that settle in less than a week. Iliso does have significant inventory, accounts payable, and accrued expenses on its balance sheet.

1. How might an analyst use Iliso's disclosure of GMV by product category?

Hide Solution

Solution:

An analyst could estimate cost of sales/sales and gross profit margin by category by looking at other retailers that primarily sell products in the specific category. Information on those companies could be used to check whether Iliso's gross profit margin is consistent with that implied by the product category mix. Also, if the mix is expected to change, an analyst can adjust the aggregate cost of sales/sales and gross profit margins accordingly. The information should probably not be used directly to develop detailed forecasts by category, because those forecasts would not be verifiable and would not be an efficient use of the analyst's time.

2. It is *most* appropriate to forecast Iliso's cost of revenue as a percentage of its:

A. retail sales.
B. total revenues.
C. total GMV sales.

Hide Solution

Solution:

not provide information on such use by third parties, and it would be a small cost compared with cost of revenue for Iliso's retail sales. Iliso's cost of revenue is primarily composed of the inventory costs of products sold to customers where Iliso records revenue gross.

3. Discuss the use of a historical base rates and convergence approach, calculated using other retailers' financial results, for forecasting Iliso's operating expenses.

Hide Solution

Solution:

Iliso's e-commerce business model and the breadth of the retail sales industry may make industry averages inappropriate to use for Iliso. However, if there are larger, more mature e-commerce retailers to calculate statistics for, this approach may be appropriate.

4. If food and grocery increases as a proportion of Iliso's retail sales, which of the following would *most likely* increase?

A. Inventory turnover
B. Gross profit margin
C. Current assets/sales

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Solution:

A is correct. Inventory turnover would increase because of the perishable nature of food and grocery. Inventory as a percentage of sales would decrease, resulting in a decrease in current assets to sales.

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CR Can someone explain the COGS calculation in Example 2 Question 2? Thanks!

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